UNIVERSITY OF BRISTOL
Electrical and Electronics Engineering Practice EEME20002

Assessment Details 2025-26

The summative assessment for this unit is composed of

A. **Coursework Portfolio (50%) :** This consists of five components that are detailed in this document.
B. **Summer Exam (50%) :** This is a two-hour, in-person written exam which tests your knowledge in the Digital Design material covered in TB1 (1 hr, 25%) and the Systems and Control material covered in TB2 (1 hr, 25%).

The remainder of this document is related to the coursework submissions for the unit.

Coursework Portfolio Details

The following components constitute your summative coursework for the unit. They are worth a combined 50% of the total unit mark.

Component	Type	Deadline
1. Project Plan	must-submit formative; Group submission	05 Dec 2025 at 13:00
2. Digital Design Practical Quiz	must-pass; Individual submission	09 Dec 2025 at 13:00
3. Personal Development Plan	must-submit; individual submission	16 April 2026 at 13:00
4. Project Review Video and Documentation	15% of mark; Group submission	16 April 2026 at 13:00
5. Digital Design Project	35% of mark; Group submission	16 April 2026 at 13:00

- *Items 1, 4 and 5 are group submissions* whereby one member of the group can submit on behalf of their own group.
- *Items 1 and 3 are must-submit: that is failure to submit will result in an automatic fail mark on the unit.*

Coursework Overview

The group design project, upon which the coursework is built, is a core component of the Electrical and Electronic Engineering Practice unit. It provides hands-on experience in digital system design using VHDL and FPGA prototyping, fostering both technical and professional skills. The project aims to simulate real-world engineering scenarios, promoting teamwork, problem-solving, and the application of digital design principles.

Coursework Learning Outcomes

On successful completion of your coursework, you will be able to:

- interpret the specification of a digital system
- structure a design into components and define component interfaces
- use VHDL to model and design digital components and systems
- approach the testing and simulation in a systematic manner
- use simulation and placement & routing tools for FPGAs
- demonstrate successful application of engineering practice skills to complete a complex project in a professional manner

Part 1 -Group: Project Plan

You are required to submit a project plan at the end of TB1 detailing how you plan to tackle the group project in term two. A template is provided and includes a Project Charter, Work Breakdown Structure, Project Timeline and Risk Register. Details and examples will be provided in the Project Management lecture series. This is a formative assessment that offers the opportunity to get feedback on your project plan before the beginning of next semester.

Part 2 – Individual: Digital Design Practical Quiz

This submission is based on a designated mandatory laboratory practical which individual students must complete and is conducted in TB1. You will be completing weekly laboratory-based tasks on Blackboard for which feedback is provided by your Team Lead.

You may access the Blackboard page for the quiz here.

Answer all questions in the quiz based on what you have carried out during TB1 and the feedback received. You are expected to achieve a minimum of 60% passing threshold for this task. *Failure to achieve the pass threshold on this practical component will result in automatic fail mark on the unit.*

Part 3 -Individual: Personal Development Plan

The Personal Development Plan is your opportunity to reflect on your personal development and to set yourself goals for the coming year and beyond. It is a mandatory submission to meet your life-long learning Intended Learning Outcome (ILO) for the unit. The form has been designed in accordance with based on the industry's UK-SPEC competencies. It can be completed and submitted at any time throughout the year. The Project Management series has put time aside for you to complete the form during a workshop session if you wish to discuss it with the academic team. The plan will not be assessed as it is personal to you and your aspirations.

Part 4 -Group: Project Review Video and Documentation

You are required to submit a 10min video reflecting on how you managed your project and contributes **15%** to the final summative mark. A slide deck template is provided, and each member must present at least one project management area:

- Project Summary and Charter
- Team Structure
- Plan
- Execution
- Outcomes
- Lessons Learned

A copy of your project management material (e.g., management workbook, meeting minutes, and plan) must also submitted as evidence to what has been reported in the video.

Marking Scheme

This will be provided to students at the start of term. Marking criteria map to the areas listed in the template and will be assessed on the 21-point scale with each area given equal weighting. The scheme applies the generic marking scale published in the University's Academic Regulations 2024-25.

Part 5 -Group: Digital Design Submission (single Zipped file)

Requirements

The entire team is required to submit a single entry through one group member. Ensure that all necessary files are included in your submission. The zip file must contain:

1. VHD code for the CMD processor and any associated submodules.
2. VHD code for the data processor and any associated submodules.
3. The generated bit file, if integration has been completed.
4. Team Technical report (as described below), including the Project Schedule and Work Division Form.

Please verify that you have submitted the final version of your project and confirm that all crucial design files are included in your submission.

Important: Please ensure that the Project Schedule and Work Division Form is the same one you submit for part 3 (Project Management Report).

Marking

This part of your coursework submission (part 2) is worth 35% of the overall unit mark distributed as follows:

1- **Team Technical Report (5%)**
2- **Submitted files (30%).**

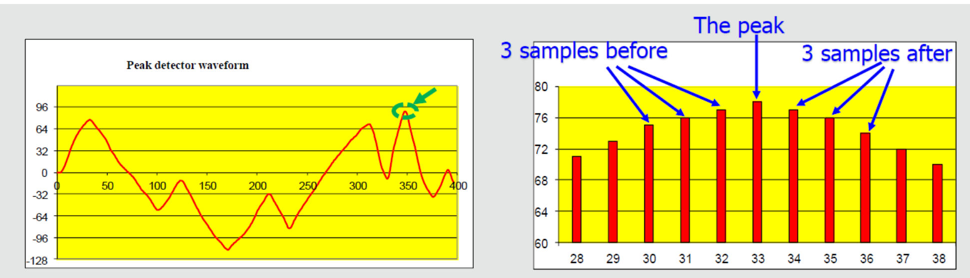
Assignment Goals

- To provide you with hands-on experience in digital design using VHDL and FPGA prototyping
- To familiarize with related industry standard tools (ModelSim, Vivado)
- To work in a team to design a digital system and implement it in an FPGA

Assignment Summary

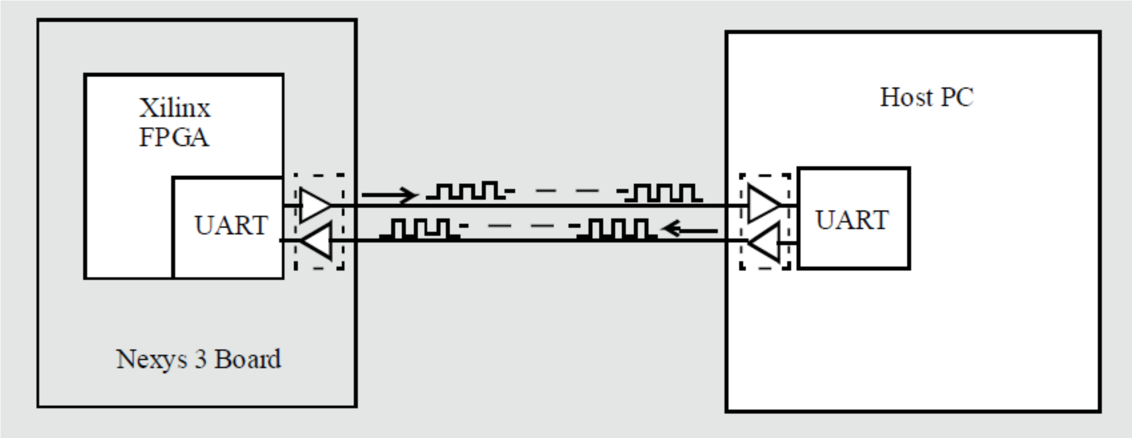
You are required to design, build and test a peak detector on an FPGA board which is under the control of a computer. The system operates via a terminal emulator, such as PuTTY, facilitating command input from an external PC. The assignment specifies using a Xilinx Artix 7 FPGA, which communicates with the external PC using the RS232 protocol. Key functionalities include processing a specified number of data bytes, detecting peak values, and returning related byte sequences. Commands allow users to start data processing, list peak results, and reset the system. The coursework emphasizes understanding and effectively utilizing the specifications for implementing the system. Key specifications are listed as follows:

- Set of user commands accepted from PC keyboard, output printed on PC screen
- Start command to process NNN bytes of data. Print each byte processed in the terminal window in hexadecimal format, separated by spaces.



- Capture of seven points around the highest peak; 3 on each side
- Print the peak byte itself, followed by a space and the peak index in decimal format.
- Use RS232 serial protocol to communicate between PC and FPGA

- Serial communication Implemented by Universal Asynchronous Receiver Transmitter (UART)



Please access more details on the technical aspects of the project on Blackboard, including the project planning document available at https://seis.bristol.ac.uk/~sy13201/digital_design/ECAD/A2_index.htm

Team Technical Report

In terms of format, this report must be written in **font size 11pt, with a maximum of 10 pages (single-spaced).**

In terms of content, the report must contain the following four components:

1. *Project Schedule and Work division Form:* Please fill in the below form and added it as the first part of your technical report. This must be the same form that you also submit for the Project Management report and is intended to provide a uniform guideline for both the technical and professional practice aspects of the project.
2. *Architecture of the Data Processor:* including a block-level sketch of the main logic components and an algorithmic state diagram with corresponding simulation results.
3. *Architecture of the Command Processor:* including a block-level sketch of the main logic components and an algorithmic state diagram with corresponding simulation results.
4. *Integration and system evaluation.*

Academic integrity

Please note that all the work that you submit, including the code, must be your group's original work. MOSS is set up at point of submission to compare submitted code against large repository including everything available on the internet, and previous years' efforts https://theory.stanford.edu/~aiken/moss/

If you use other author's code that you have obtained, you MUST EXPLICITLY label it as such and provide a reference to avoid issue of plagiarism. For example, "the function Y was obtained from [GitHub page Z]. (link here)"

I*f you are unsure on what constitutes plagiarism, please consult the University of Bristol guidance on Plagiarism on Blackboard. See also*
https://www.bristol.ac.uk/students/support/academic-advice/academic-integrity/plagiarism/

Marking Scheme

This will be provided to students at the start of term. The scheme will be derived from the generic marking scale published in the University's Academic Regulations 2025-26

Important: In preparing your coursework, please note that the use of AI tools is classified as category 2 (minimal) as explained in the guidelines on use of AI in assignments at the University of Bristol. Please make sure you read what this means at the following link https://www.bristol.ac.uk/bilt/sharing-practice/guides/guidance-on-ai/using-ai-in-assessment/